

STUDENT NAME



YEAR 12 Mathematics

HSC Course

Assessment Task 3

2010

Time Allowed – 70 Minutes

1. There are 4 questions, NOT of equal value.
2. Answer each question on your own paper.
3. Start each Question on a new sheet of paper.
4. Show all necessary working
5. Use one side of the paper only.
6. Calculators may be used

Topic	Mark
1. Question 1 (Trigonometric Functions)	/16
2. Question 2 (Logs and Exponentials)	/17
3. Question 3 (Trigonometric Functions)	/18
4. Question 4 (Logs and Exponentials)	/16
Total	/67

Question 3 (18 Marks) (Start a New Sheet of Paper)**Marks**

- a) Find the equation of the normal to the curve $y = 5 \sin 2x$ at the point where $x = \frac{\pi}{6}$.
Do not simplify your answer. 3
- b) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ 2
- c) For $y = 4 \cos 3x - 2 \sin 3x$, show that $\frac{d^2 y}{dx^2} = -9y$ 2
- d) (i) Show that $x = \frac{\pi}{3}$ is a solution to the equation $\sin 2x = \sin x$ 1
- (ii) Sketch on the same axes, the curves $y = \sin 2x$ and $y = \sin x$ for $0 \leq x \leq 2\pi$. 2
- (iii) Write down 2 more solutions to the equation $\sin 2x = \sin x$ in this domain. 2
- (iv) Find the area between $y = \sin 2x$ and $y = \sin x$ for $\frac{\pi}{3} \leq x \leq \pi$ 3
- e) Use Simpson's Rule with 5 function values to approximate $\int_0^{\pi} \sin^2 x \, dx$. 3

Question 4 (16 Marks) Start a New Sheet of Paper**Marks**

- a) (i) Differentiate xe^{2x} 2
- (ii) Hence find $\int xe^{2x} dx$ 3
- c) For the curve $y = \frac{\ln x}{x}$
- (i) state the domain of the function 1
- (ii) find the coordinates of the stationary point 2
- (iii) establish the nature of the stationary point 1
- Given that there is a point of inflexion when $x = e^{\frac{3}{2}}$ (no need to find y)
- (iv) sketch the curve showing any intercepts with the axes 2
- d) (i) Sketch the curve $y = \log_e(1+x)$ from $x=0$ to $x=1$ 1
- (ii) The region bounded by this portion of the curve and the y axis is rotated about the y axis. Find the volume of the solid of revolution so formed. 4

Question 1 (16 Marks)		Marks
a)	Find the exact value of $\sec \frac{5\pi}{6}$	2
b)	A chord in a circle with radius 8 cm subtends an angle of 60° at the centre. Find the exact area of the minor segment cut off by the chord.	3
c)	Solve $2 \cos 2x = -1$ for $0 \leq x \leq 2\pi$	4
d)	Differentiate:	
(i)	$x \cos 3x$	2
(ii)	$\frac{5}{\sin^3 x}$	2
e)	Evaluate exactly $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \sec^2 \frac{x}{2} dx$	3
Question 2 (17 Marks) Start a New Sheet of paper		Marks
a)	Evaluate exactly $\log_2 \sqrt{8}$	2
b)	Solve to 3 significant figures $2^x = 27$	2
c)	Simplify $\frac{\log_e k}{\log_e a} + \log_a k^2$	2
d)	Differentiate:	
(i)	e^{-3x}	1
(ii)	$\ln(5x - 3)$	1
(iii)	$\log_e \sqrt{\frac{3x-2}{2x+1}}$	3
e)	Find $\int \frac{4x^3 - 5x}{x^2} dx$	3
f)	Evaluate in simplest form $\int_2^3 \frac{x}{x^2 + 1} dx$	3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

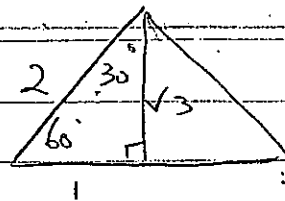
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

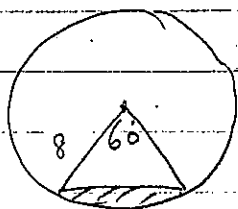
2010 Mathematics 45C Task 3

Question 1

$$\begin{aligned} \text{a) } \sec \frac{5\pi}{6} &= \sec 150^\circ \\ &= -\sec 30^\circ \\ &= -\frac{2}{\sqrt{3}} \end{aligned}$$



b)



$$\begin{aligned} A &= \frac{1}{2} r^2 (\theta - \sin \theta) \\ &= \frac{1}{2} \times 8^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right) \\ &= 32 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) \text{ cm}^2 \end{aligned}$$

c)

$$\cos 2x = -\frac{1}{2}$$

$$0 \leq 2x \leq 4\pi$$

$2x$ lies in 2nd or 3rd quad.

Acute angle is $\frac{\pi}{3}$

$$\therefore 2x = \pi - \frac{\pi}{3} \text{ or } \pi + \frac{\pi}{3} \text{ or } 3\pi - \frac{\pi}{3} \text{ or } 3\pi + \frac{\pi}{3}$$

$$= \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} \text{ or } \frac{8\pi}{3} \text{ or } \frac{10\pi}{3}$$

$$x = \frac{\pi}{3} \text{ or } \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} \text{ or } \frac{5\pi}{3}$$

d)

$$\text{i) } y = x \cos x$$

$$y' = -3x \sin 3x + \cos 3x$$

ii)

$$y = 5(\sin x)^{-3}$$

$$y' = -15(\sin x)^{-4} \cos x$$

$$= \frac{-15 \cos x}{\sin^4 x}$$

e)

$$\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \sec^2 \frac{x}{2} dx = \left[2 \tan \frac{x}{2} \right]_{\frac{\pi}{3}}^{\frac{2\pi}{3}}$$

$$= 2 \left\{ \tan \frac{\pi}{3} - \tan \frac{\pi}{6} \right\}$$

$$= 2 \left\{ \sqrt{3} - \frac{1}{\sqrt{3}} \right\}$$

$$= 2 \left\{ \frac{3-1}{\sqrt{3}} \right\} = \frac{4}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{4\sqrt{3}}{3}$$

Question 2

$$a) \text{ Let } \log_2 \sqrt{8} = u \quad \therefore 2^u = \sqrt{8} = 2^{3/2}$$
$$\therefore u = \frac{3}{2}$$

$$\therefore \log_2 \sqrt{8} = \frac{3}{2}$$

$$b) \quad x \ln 2 = \ln 27$$

$$x = \frac{\ln 27}{\ln 2} = 4.7548875$$

$$x \doteq 4.75$$

$$c) \quad \log_a k + 2 \log_a k = 3 \log_a k$$

$$d) \quad i) \quad y' = -3e^{-3x}$$

$$ii) \quad y' = \frac{5}{5x-3}$$

$$iii) \quad y = \frac{1}{2} \{ \ln(3x-2) - \ln(2x+1) \}$$

$$y' = \frac{3}{2(3x-2)} - \frac{1}{2x+1}$$

$$e) \quad = \frac{e^{-3x}}{3} + C$$

$$f) \quad \int 4x - \frac{5}{x} dx = 2x^2 - 5 \ln x + C$$

$$g) \quad \frac{1}{2} \int_2^3 \frac{2x dx}{x^2+1} = \frac{1}{2} \left[\ln(x^2+1) \right]_2^3$$

$$= \frac{1}{2} (\ln 10 - \ln 5)$$

$$= \frac{1}{2} \ln \frac{10}{5}$$

$$= \frac{1}{2} \ln 2$$

Question 3

a) $y = 5 \sin 2x$ $x = \frac{\pi}{6}$, $y = 5 \sin \frac{\pi}{3} = \frac{5\sqrt{3}}{2}$
 $\left(\frac{\pi}{6}, \frac{5\sqrt{3}}{2}\right)$

$y' = 10 \cos 2x$ $x = \frac{\pi}{6}$ $m_1 = 10 \cos \frac{\pi}{3} = 5$

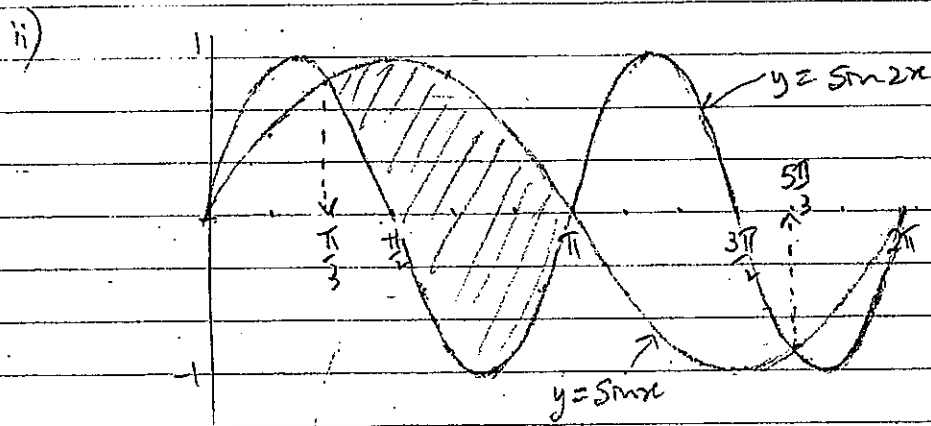
$\therefore m_2 = -\frac{1}{5}$

$\therefore y - \frac{5\sqrt{3}}{2} = -\frac{1}{5} \left(x - \frac{\pi}{6}\right)$

b) $3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = 3 \times 1 = 3$

c) $y' = -12 \sin 3x - 6 \cos 3x$
 $y'' = -36 \cos 3x + 18 \sin 3x$
 $= -9(4 \cos 3x - 2 \sin 3x)$
 $= -9y$

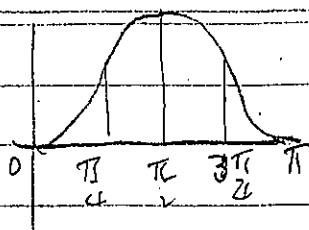
d) i) $\sin 2 \times \frac{\pi}{3} = \sin 120^\circ = +\sin 60^\circ = \frac{\sqrt{3}}{2}$
 $\sin \frac{\pi}{3} = \sin 60^\circ = \frac{\sqrt{3}}{2}$
 $\therefore x = \frac{\pi}{3}$ is a solution



iii) Other solutions are $x = \pi$ & $x = 2\pi - \frac{\pi}{3}$ i.e. $x = \frac{5\pi}{3}$

iv) $\text{Area} = \int_{\pi/3}^{\pi} \sin x - \sin 2x \, dx = \left[-\cos x + \frac{1}{2} \cos 2x \right]_{\pi/3}^{\pi}$
 $= -\cos \pi + \frac{1}{2} \cos \frac{\pi}{3} + \cos \frac{\pi}{3} - \frac{1}{2} \cos \frac{2\pi}{3}$
 $= -(-1) + \frac{1}{2} \left(\frac{1}{2}\right) + \frac{1}{2} - \frac{1}{2} \left(-\frac{1}{2}\right)$
 $= 1 + \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 2$

c)



$$h = \frac{\pi}{4}$$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$y(x)$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0

$$I \doteq \frac{h}{3} \{y_0 + 4y_1 + y_2\} + \frac{h}{3} \{y_2 + 4y_3 + y_4\}$$

$$I \doteq \frac{h}{3} \{y_0 + y_4 + 4(y_1 + y_3) + 2y_2\}$$

$$I \doteq \frac{\pi}{12} \{0 + 0 + 4(\frac{1}{2} + \frac{1}{2}) + 2 \times 1\}$$

$$I \doteq \frac{\pi}{12} \times 6$$

$$I \doteq \frac{\pi}{2}$$

Question 4

i) $y = x e^{2x}$
 $y' = x \cdot 2e^{2x} + e^{2x} \cdot 1$

$$y' = 2x e^{2x} + e^{2x}$$

ii)

$$\therefore \int 2x e^{2x} + e^{2x} dx = x e^{2x}$$

$$2 \int x e^{2x} dx + \int e^{2x} dx = x e^{2x}$$

$$2 \int x e^{2x} dx = x e^{2x} - \int e^{2x} dx$$

$$= x e^{2x} - \frac{e^{2x}}{2}$$

$$\int x e^{2x} dx = \frac{x}{2} e^{2x} - \frac{e^{2x}}{4} + C$$

c) $y = \frac{\ln x}{x}$

i) $D: x > 0$ (for $\ln x$ to exist)

ii) $y' = \frac{x \cdot \frac{1}{x} - \ln x}{x^2}$
 $= \frac{1 - \ln x}{x^2}$

For Stat pt $y' = 0$ $1 - \ln x = 0$

$\ln x = 1$

$x = e$

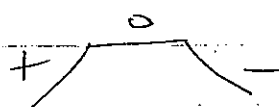
$y = \frac{1}{e}$

Stat pt at $(e, \frac{1}{e})$

iii)

x	e^-	e	e^+
y'	$+$	0	$-$

using y'



$(e, \frac{1}{e})$ is a maximum turning point

iv) For points of inflexion $y'' = 0$

$y'' = \frac{x^2 \left(-\frac{1}{x}\right) - (1 - \ln x) \cdot 2x}{x^4}$

$= \frac{-x - 2x + 2x \ln x}{x^4}$

$= \frac{-3x + 2x \ln x}{x^4} = \frac{-3 + 2 \ln x}{x^3}$

pt of inflection when

$$-3 + 2 \ln x = 0$$

$$2 \ln x = 3$$

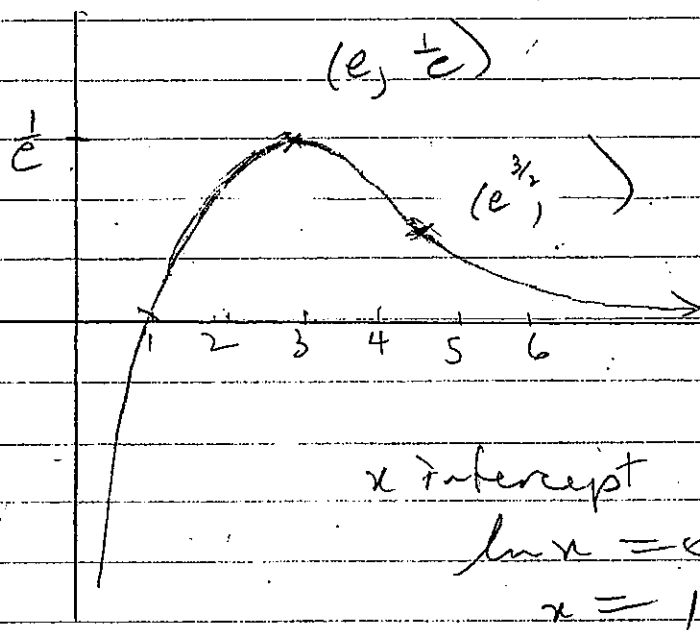
$$\ln x = \frac{3}{2}$$

$$x = e^{3/2}$$

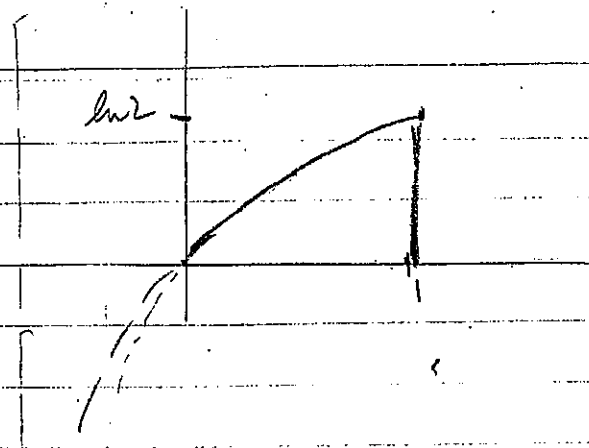
Show y'' changes sign

x	$e^{3/2}$	$e^{3/2}$	$e^{3/2}$
y''	-	0	+

Concavity changes from concave down to concave up



2)



$$y = \ln(1+x)$$

$$1+x = e^y$$

$$x = e^y - 1$$

$$x^2 = e^{2y} - 2e^y + 1$$

$$V = \pi \int x^2 dy$$

$$= \pi \int_0^{\ln 2} (e^{2y} - 2e^y + 1) dy$$

$$= \pi \left[\frac{e^{2y}}{2} - 2e^y + y \right]_0^{\ln 2}$$

$$= \pi \left(\frac{e^{2\ln 2}}{2} - 2e^{\ln 2} + \ln 2 \right) - \left(\frac{1}{2} - 2 + 0 \right)$$

$$= \pi \left(\frac{4}{2} - 2 \times 2 + \ln 2 \right) - \left(-\frac{3}{2} \right)$$

$$= \pi \left[2 - 4 + \ln 2 + \frac{3}{2} \right]$$

$$= \pi \left(\ln 2 - \frac{1}{2} \right) \pi^3 = \frac{\pi}{2} (2\ln 2 - 1)$$